

Global Variation in Lipoprotein(a) Levels Among Patients With Coronary Heart Disease



Insights From the INTERASPIRE Study and Implications for Emerging Lp(a)-Lowering Therapies

Fotios Barkas, MD, PhD,^{a,b} Julia Brandts, MD,^{a,c} Dirk De Bacquer, PhD,^d Catriona Jennings, PhD,^e Guy G. De Backer, MD, PhD,^d Kornelia Kotseva, MD, PhD,^{e,f} Lars Ryden, MD, PhD,^g Gregory Y.H. Lip, MD,^h Raul D. Santos, MD, PhD,^{i,j} Peter Libby, MD,^k Iris Erlund, PhD,^l Sandra Ganly, PhD,^e Terhi Vihervaara, PhD,^l Agnieszka Adamska, MBA, MSc,^e Ana Abreu, MD, PhD,^{m,n} Wael Almahmeed, MD,^o Ade Meidian Ambari, MD,^p Junbo Ge, MD, PhD,^q Hosam Hasan-Ali, MD, PhD,^r Yong Huo, MD,^s Piotr Jankowski, MD, PhD,^t Rodney M. Jimenez, MD,^u Yong Li, MD, PhD,^v Ahmad Syadi Mahmood Zuhdi, MD,^w Abel Makubi, MD,^x Amam Chinyere Mbakwem, MBBS,^y Lilian Mbau, MD,^z Jose Luis Navarro Estrada, MD,^{aa} Okechukwu Samuel Ogah, MBBS, PhD,^{bb} Elijah Nyainda Ogola, MBChB, MMEd,^{cc} Adalberto Quintero-Baiz, MD,^{dd} Mahmoud Umar Sani, MBBS, PhD,^{ee} Maria Ines Sosa Liprandi, MD,^{ff} Jack Wei Chieh Tan, MBBS,^{gg,hh,ii} Miguel Alberto Urina Triana, MD,^{jj} Tee Joo Yeo, MBBS,^{kk} David Wood, MB ChB, MSc,^{e,ll} John William McEvoy, MB, BCh, BAO, MEd, MHS, PhD,^e Kausik K. Ray, MD, FMedSci,^a the INTERASPIRE Investigators

ABSTRACT

BACKGROUND Lipoprotein(a) [Lp(a)] is a common risk factor for atherosclerotic cardiovascular disease, potentially more atherogenic per particle than low-density lipoprotein. An estimated 1.5 billion individuals globally have elevated levels ≥ 125 nmol/L, considered as a risk-enhancing threshold. Although Lp(a) levels vary by ethnicity, ongoing trials of novel therapies in predominantly secondary prevention patients use fixed Lp(a) enrollment thresholds.

OBJECTIVES The purpose of this study was to assess Lp(a) levels in coronary heart disease (CHD) patients across geographical regions, providing inference on the proportions potentially eligible for future Lp(a)-lowering therapies and whether these vary by region and country.

METHODS INTERASPIRE (International Action on Secondary Prevention through Intervention to Reduce Events) enrolled adults hospitalized with CHD in the previous 6 to 24 months. Lp(a) levels were available in 13 countries across 6 World Health Organization (WHO) regions: Africa (Kenya, Nigeria, Tanzania), Americas (Argentina, Colombia), Eastern Mediterranean (UAE), Europe (Poland, Portugal), South-East Asia (Indonesia), and Western Pacific (China, Malaysia, Philippines, Singapore). Lp(a) measurements were performed once and centrally in Helsinki using an isoform-independent assay for 11 countries, and locally in Indonesia and China with standardization to the core laboratory. Lp(a) levels are reported as median (Q1-Q3) and proportions above different thresholds.

RESULTS Lp(a) results were available for 3,928 patients from 13 countries (mean age: 60.2 ± 10.2 years; 21.1% women). Median Lp(a) was 32 nmol/L (Q1-Q3: 11-89 nmol/L) overall, with 17.6% having levels ≥ 125 nmol/L. Median levels varied by region—highest in Africa (62 nmol/L) and lowest in Western Pacific (22 nmol/L)—and also between countries within regions: Europe (Portugal: 59 nmol/L vs Poland: 19.5 nmol/L), South America (Colombia: 46 nmol/L vs Argentina: 32 nmol/L) and Western Pacific (Malaysia: 39.5 nmol/L vs Philippines: 14 nmol/L). Overall, the proportions of patients with Lp(a) ≥ 150 , 175, and 200 nmol/L (hence eligibility for future Lp(a)-lowering therapies) were 13.0%, 9.3%, and 6.2%, respectively, with eligibility also varying among countries: highest in Portugal (25.5%, 18.3%, and 11.6%) and lowest in Philippines (4.3%, 2.5%, and 1.3%).

CONCLUSIONS The vast majority of patients with CHD have Lp(a) levels far below what is considered a typical risk-enhancing threshold, suggesting that the attributable risk from Lp(a) is more complex than previously perceived. Furthermore, wide geographical variations in Lp(a) levels above entry criteria for ongoing trials could impact equitable access to therapies, if these trials are positive. (JACC. 2025;85:2028-2042) © 2025 by the American College of Cardiology Foundation.



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Lipoprotein(a) [Lp(a)] is an apolipoprotein (apo) B100-containing lipoprotein covalently bound to a unique glycoprotein produced in the liver called apo(a).^{1–3} Genetic variations in apo(a), which account for 70% to 90% of the interindividual variability in Lp(a) levels, and ethnicity (where Black individuals typically have the highest levels and Chinese populations the lowest), contribute to the known variability in Lp(a) levels in different regions of the world.^{1,2,4} Epidemiological studies estimate that 1.5 billion individuals worldwide have elevated Lp(a) levels, with ≥ 125 nmol/L typically considered a threshold associated with heightened risk of atherosclerotic cardiovascular disease (ASCVD).^{1,2} Although the relationship between Lp(a) and ASCVD remains log-linear irrespective of the distribution of Lp(a) within different ethnic groups,^{1,2} recent data suggest that the population-attributable risk from specific Lp(a) levels may vary by ethnicity and that the Lp(a) thresholds for cardiovascular disease events may vary between primary and secondary prevention populations.^{5,6}

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Although Lp(a) particles carry less cholesterol than other ApoB-containing lipoproteins such as low-density lipoprotein (LDL), Lp(a) may confer a higher atherogenic risk per ApoB particle compared with LDL.^{7–10} These observations, when taken together with a lack of currently available therapies capable of providing meaningful reductions in Lp(a), have led to speculation that high Lp(a) may contribute to the

global burden of ASCVD much more than other conventional risk factors.¹¹ In this context, ongoing cardiovascular outcome trials are evaluating novel Lp(a)-lowering therapies in patients primarily with ASCVD and high Lp(a) levels ranging from ≥ 150 ¹² to ≥ 200 nmol/L.¹³ Should these trials provide evidence of cardiovascular benefit, access to these different therapies in different geographical regions may vary widely given the known variability in Lp(a) by region, resulting in potential inequalities across different parts of the world. To provide further insights into these considerations, we measured Lp(a) in patients participating in the INTERASPIRE (International Action on Secondary Prevention through Intervention to Reduce Events) study, an international study evaluating implementation and attainment of guideline standards for secondary prevention and cardiac rehabilitation in patients with a coronary event in the previous 6 to 24 months.

METHODS

STUDY POPULATION. The design of the INTERASPIRE study has been reported elsewhere.^{14,15} In brief, INTERASPIRE was a cohort study aiming to examine guideline-recommended risk factor control in patients with a recent diagnosis of coronary heart

ABBREVIATIONS AND ACRONYMS

apo = apolipoprotein
ASCVD = atherosclerotic cardiovascular disease
BMI = body mass index
BP = blood pressure
CABG = coronary artery bypass graft surgery
CHD = coronary heart disease
CKD = chronic kidney disease
eGFR = estimated glomerular filtration rate
HbA1c = glycated hemoglobin
HDL-C = high-density lipoprotein cholesterol
LDL-C = low-density lipoprotein cholesterol
LLM = lipid-lowering medication
Lp(a) = lipoprotein(a)
MI = myocardial infarction
non-HDL-C = non-high-density lipoprotein cholesterol
PAD = peripheral artery disease
PCI = percutaneous coronary intervention
TC = total cholesterol
TG = triglyceride
VLDL-C = very-low-density lipoprotein cholesterol

From the ^aDepartment of Public Health and Primary Care, Imperial College London, United Kingdom; ^bFaculty of Medicine, School of Health Sciences, University of Ioannina, Ioannina, Greece; ^cDepartment of Internal Medicine-Cardiology, RWTH Aachen University, Aachen, Germany; ^dDepartment of Public Health and Primary Care, Ghent University, Ghent, Belgium; ^eUniversity of Galway School of Medicine and National Institute for Prevention and Cardiovascular Health, Moyola Lane, Newcastle, Galway, Ireland; ^fImperial College Healthcare NHS Trust, Charing Cross Hospital, Fulham Palace Road, London, United Kingdom; ^gDepartment of Medicine Solna, Karolinska Institutet, Stockholm, Sweden; ^hLiverpool Centre for Cardiovascular Science at University of Liverpool, Liverpool John Moores University and Liverpool Heart and Chest Hospital, Liverpool, United Kingdom; ⁱHospital Israelita Albert Einstein, São Paulo, Brazil; ^jHeart Institute (InCor) University of São Paulo Medical School Hospital, São Paulo, Brazil; ^kDivision of Cardiovascular Medicine, Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, Massachusetts, USA; ^lFinnish Institute for Health and Welfare, Helsinki, Finland and Institute for Nutrition and Health Research, Helsinki, Finland; ^mCardiology Service, Cardiovascular Rehabilitation Unit, Hospital S. Maria (ULSSM), Lisbon, Portugal; ⁿFaculty of Medicine University of Lisbon, CAML, ISAMB, IMPSP, CCUL, Lisbon, Portugal; ^oCleveland Clinic Abu Dhabi, Al Mariah Island, Abu Dhabi, United Arab Emirates; ^pNational Cardiovascular Center Harapan Kita, Department of Cardiology and Vascular Medicine, Faculty of Medicine, Universitas Indonesia, West Jakarta, Jakarta, Indonesia; ^qFudan University Zhongshan Hospital, Xuhui District, Shanghai, China; ^rCardiovascular Medicine Department, Assiut University Heart Hospital, Faculty of Medicine, Assiut University, Assiut, Egypt; ^sPeking University First Hospital, Department of Cardiology, Institute of Cardiovascular Disease, Beijing, China; ^tDepartment of Internal Medicine and Geriatric Cardiology, Centre of Postgraduate Medical Education, Warsaw, Poland; ^uPhilippine Heart Association and St Luke's Medical Center, Global City, Taguig City, Metro Manila, Philippines; ^vDepartment of Internal Medicine (Cardiology), Fudan University Huashan Hospital, Shanghai, China; ^wUniversity Malaya Medical Centre, Cardiology Unit, Department of Medicine, Kuala Lumpur, Malaysia; ^xMuhimbili University of Health and Allied Sciences, Dar es Salaam, Tanzania; ^yCollege of Medicine, University of Lagos, Idi Araba, Lagos, Nigeria; ^zKenya Cardiac Society, Nairobi, Kenya; ^{aa}Cardiology Division, Italian Hospital of Buenos Aires, Buenos Aires, Argentina; ^{bb}Cardiology Unit, Department of Medicine, University of Ibadan, University College Hospital, Ibadan, Nigeria; ^{cc}Department of Clinical Medicine and Therapeutics, University of Nairobi, Nairobi, Kenya; ^{dd}Faculty of Health Sciences, University Simon Bolívar,

disease (CHD) and conducted in 14 countries across 6 WHO regions: Africa (Kenya, Nigeria, and Tanzania), the Americas (Argentina and Colombia), the Eastern Mediterranean Region (Egypt and UAE), Europe (Poland and Portugal), South-East Asia (Indonesia), and the Western Pacific Region (China, Malaysia, Philippines, and Singapore). For the present analysis, data from Egypt were excluded because Lp(a) measurements were not performed, resulting in 13 countries being included.

Eligible patients aged 18 to 79 years at their index hospitalization for CHD (acute myocardial infarction [MI], elective or emergency percutaneous coronary intervention [PCI], coronary artery bypass graft surgery [CABG], or unstable angina) were identified retrospectively from hospital records across selected hospitals. Patients were followed up between 6 months and 2 years post-index hospitalization (and up to 3 years in Africa to facilitate recruitment) for a standardized interview and examination. Ethics approvals were secured from each hospital, and written informed consent was obtained from all participants. The study aimed to enroll 400 coronary patients per country.

ETHICAL PROCEDURES. National coordinators were responsible for obtaining the Local Research Ethics Committees' approvals. Written informed consent was obtained from each participant by the investigator using a signed declaration.

STUDY PROCEDURES. Data were collected at interview by centrally trained researchers using standardized methods for all interviews, examinations, and laboratory measurements, including patient demographics, medical history, health behaviors, and medication use. Standardized procedures for height, weight, carbon monoxide in breath, and blood pressure (BP) measurements were applied as previously described in detail elsewhere.^{14,15} These were collected electronically using a customized database equipped with range and consistency checks and submitted online to data management centers at the EURObservational Research Programme, European Heart House, Sophia Antipolis, France, and ARO

Specialist Data Centre Services (previously AIMES), Liverpool, United Kingdom. The data were reviewed for completeness, internal consistency, and accuracy and stored in compliance with national data protection regulation.

LABORATORY MEASUREMENTS. Venous fasting blood samples were collected and serum and plasma samples were aliquoted and stored at -70°C . All frozen samples, except those from China and Indonesia, were transported on dry ice to a Central Laboratory, Finnish Institute for Health and Welfare (THL), accredited to SFS-EN ISO/IEC 17025:2005 standards. Centralized analyses included serum total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), triglyceride (TG), ApoB, Lp(a), creatinine, fasting plasma glucose, and whole blood glycated hemoglobin (HbA1c). All central laboratory measurements were performed on a clinical chemistry analyzer (Architect c8000, Abbott Laboratories), which used immunoturbidimetric assays for the measurement of ApoB (Abbott) and isoform-independent immunoassays for Lp(a) (Randox). Samples from China and Indonesia were analyzed in local accredited laboratories standardized to the central laboratory, where the Roche Cobas Tinaquant assay was used for Lp(a) measurement. ApoB levels were not available in patients from these 2 countries.

Low-density lipoprotein cholesterol (LDL-C) levels were calculated using the Sampson formula, optimizing accuracy in patients with low LDL-C or hypertriglyceridemia ($\text{TG} \leq 800 \text{ mg/dL}$): $\text{LDL-C}_{\text{Sampson}} = \text{TC} / 0.948 - \text{HDL-C} / 0.971 - (\text{TG} / 8.56 + [\text{TG} \times \text{non-HDL-C}] / 2140 - \text{TG}^2 / 16,100) - 9.44$.¹⁶ The Sampson formula uses mg/dL; therefore, conversions were required for measurements expressed in mmol/L. To convert TG levels from mg/dL to mmol/L, we multiplied by 0.0113, and for cholesterol levels by 0.0259. Non-high-density lipoprotein cholesterol (non-HDL-C) was calculated as $\text{non-HDL-C} = \text{TC} - \text{HDL-C}$. Very-low-density lipoprotein cholesterol (VLDL-C) was estimated as $\text{VLDL-C} = \text{non-HDL-C} - \text{LDL-C}$. Glomerular filtration rate (eGFR) was estimated using the race-free

Barranquilla, Colombia; ^{ee}Department of Medicine, Bayero University Kano and Aminu Kano Teaching Hospital, Kano State, Nigeria; ^{ff}Department of Cardiology, University Hospital Sanatorio Güemes, Buenos Aires, Argentina; ^{gg}National Heart Centre Singapore, Singapore; ^{hh}Department of Cardiology, Duke-NUS Medical School, Singapore; ⁱⁱDepartment of Cardiology, Sengkang General Hospital, Singapore; ^{jj}Faculty of Health Sciences, Universidad Simon Bolivar, Barranquilla, Colombia; ^{kk}National University Heart Centre Singapore and National University Hospital, National University Health System (NUHS) Tower Block, Cardiac Department, Singapore; and the ^{ll}National Heart and Lung Institute, Imperial College London, London, United Kingdom. The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

definition of Chronic Kidney Disease (CKD) Epidemiology Collaboration (CKD-EPI) equation.¹⁷

OUTCOME MEASURES. The primary outcome of this cross-sectional analysis was to describe the distribution of Lp(a) levels overall, by region and country, along with the proportions of individuals exceeding specific Lp(a) thresholds, which align with the ASCVD risk-enhancing threshold and inclusion criteria for ongoing cardiovascular outcome trials of Lp(a)-lowering therapies. Because the Lp(a) distribution is strongly skewed, concentrations are summarized as median (Q1-Q3). The ASCVD risk-enhancing threshold was 125 nmol/L.¹ The thresholds for ongoing trials were: 150 nmol/L (70 mg/dL) for Lp(a) HORIZON (Assessing the Impact of Lipoprotein(a) Lowering with TQJ230 on Major Cardiovascular Events in Patients With CVD; [NCT04023552](#)),¹² 175 nmol/L for ACCLAIM-Lp(a) (A Study to Investigate the Effect of Lepodisiran on the Reduction of Major Adverse Cardiovascular Events in Adults With Elevated Lipoprotein(a); [NCT06292013](#)),¹⁸ and 200 nmol/L for OCEAN(a) OUTCOMES (Olpasiran Trials of Cardiovascular Events and Lipoprotein(a) Reduction-Outcomes Trial; [NCT05581303](#)).¹³

Lp(a) levels were also compared across subgroups defined by sex; age (≥ 60 years vs < 60 years); family history of premature ASCVD (defined as ASCVD in a first-degree relative before age 55 years for men or 65 years for women); concomitant diseases (self-reported), such as familial hypercholesterolemia and diabetes; and patients' eGFR. Additional comparisons were made based on the phenotype of ASCVD (MI, PCI, CABG, stroke, peripheral artery disease [PAD], poly-vascular disease, defined as the presence of atherosclerosis in more than one arterial bed), event classification (first or recurrent event), time since index CHD hospitalization (< 1 or ≥ 1 year), and concurrent lipid-lowering therapy.

To assess whether elevated Lp(a) levels can be inferred from traditional lipids and lipoproteins, we divided the overall cohort into deciles of Lp(a). Across Lp(a) deciles, we assessed LDL-C, VLDL-C, and non-HDL-C (all expressed in mmol/L); total ApoB mass (expressed in g/L); and the contribution of ApoB-containing Lp(a) particles to overall ApoB-containing particles (expressed in %). Given that ApoB and apo(a) are in a 1:1 molar relationship, Lp(a)-ApoB molar concentration was determined directly from the measured concentration of Lp(a) in nmol/L. Total ApoB in nmol/L was calculated by converting ApoB mass (g/L) to nmol/L using a factor of 19.49 (based on ApoB molecular weight of 513 kDa)¹⁹: total ApoB in nmol/L = (total ApoB (g/L) $\times$ 100) $\times$ 19.49.

Thus, the proportion of Lp(a) particles to total ApoB-containing particles was estimated as follows: Lp(a)/(total ApoB in nmol/L).

In the absence of Lp(a)-lowering therapies, one approach to mitigate the risk from elevated Lp(a) is to more intensively control traditional risk factors.¹ Participants and physicians were unaware of the results of their central core laboratory-measured Lp(a) levels for this study. Hence, to provide an understanding of how much improvement in traditional risk factors might be needed to help offset some of the ASCVD risk in patients with Lp(a) ≥ 125 nmol/L, we assessed the proportions of patients with inadequate control of lifestyle and risk factors. These included smoking (validated by breath carbon monoxide > 10 ppm), sedentary behavior (not performing moderate-to-vigorous exercise for at least 30 minutes on average 5 times per week) and thresholds indicating significantly elevated traditional risk factors: body mass index (BMI) ≥ 25 kg/m², systolic/diastolic BP $\geq 130/80$ mm Hg, LDL-C ≥ 1.4 mmol/L, and for participants with known diabetes, HbA1c $\geq 7\%$ (53 mmol/mol).^{20,21}

STATISTICAL ANALYSIS. Data were summarized overall, at region or country level, and within specific subgroups, as previously described. Continuous variables were reported as mean $\pm$ SD or median (Q1-Q3) for skewed distributions. Categorical variables were summarized as counts and percentages. Analyses included also visualization of Lp(a) distribution across countries using violin plots to assess density and spread, with Box plots providing additional detail. Cumulative distribution plots illustrated the proportion of participants exceeding predefined Lp(a) thresholds. Associations between increasing Lp(a) and other lipid-related measures across Lp(a) deciles were explored and visualized using locally estimated scatterplot smoothing plots. Lp(a) distributions were compared between subgroups according to the Kruskal-Wallis test. ORs with 95% CIs were calculated to assess the association between sex and elevated Lp(a) levels (≥ 125 nmol/L), both in crude and multivariable-adjusted models. A type I error level of $\alpha = 0.05$ was used to indicate statistical significance. All data analyses were undertaken using SAS statistical software version 9.4 (SAS Institute Inc).

RESULTS

Of eligible patients identified at each site, the overall interview rate was 39.7%, after excluding those who had moved or died or who were not able to attend because of a change in health status.¹⁴ Of the 4,548 CHD patients interviewed in INTERASPIRE, 3,928

TABLE 1 Study Participants' Characteristics Overall and By Region and Country

	Overall (N = 3,928)	AFR			AM		EM
		Kenya (n = 124)	Nigeria (n = 223)	Tanzania (n = 169)	Argentina (n = 187)	Colombia (n = 392)	UAE (n = 398)
Women	21.1 (828/3,928)	30.6 (38/124)	36.3 (81/223)	49.7 (84/169)	10.7 (20/187)	28.1 (110/392)	13.8 (55/398)
Age, y	60.2 ± 10.2	61.8 ± 10.1	58.3 ± 11.8	63.1 ± 9.7	63.3 ± 9.9	62.4 ± 9.5	56.7 ± 11
FH	9.7 (343/3,526)	1.9 (2/103)	4.5 (9/198)	5.6 (7/125)	6.7 (12/180)	5.9 (22/376)	21.1 (70/332)
Diabetes	44.0 (1728/3,928)	67.7 (84/124)	39.5 (88/223)	36.7 (62/169)	39.6 (74/187)	33.4 (131/392)	69.3 (276/398)
CKD	9.5 (368/3,863)	6.5 (8/124)	6.9 (15/218)	4.2 (7/166)	10.2 (19/187)	10.4 (40/384)	8.9 (35/393)
Lp(a), nmol/L ^a	32 (11-89)	63.5 (24.5-124)	65 (30-121)	60 (34-107)	32 (11-99)	46 (15-122)	27 (13-62)
LDL-C, mmol/L	2.28 ± 1.02	2.35 ± 1.08	2.68 ± 1.18	2.86 ± 1.08	2.13 ± 0.84	2.34 ± 0.95	2.19 ± 1.02
Non-HDL-C, mmol/L	2.95 ± 1.19	2.94 ± 1.15	3.25 ± 1.32	3.55 ± 1.23	2.88 ± 1.06	3.09 ± 1.13	2.84 ± 1.15
Total ApoB, g/L	0.83 ± 0.29	0.85 ± 0.29	0.91 ± 0.34	0.98 ± 0.33	0.81 ± 0.24	0.87 ± 0.27	0.78 ± 0.27
Low/moderate-intensity statin monotherapy	23.1 (905/3,912)	12.1 (15/124)	23.8 (53/223)	58.0 (98/169)	11.8 (22/187)	5.2 (20/384)	14.4 (57/396)
High-intensity statin monotherapy	50.7 (1,982/3,912)	71.8 (89/124)	43.5 (97/223)	5.3 (9/169)	73.8 (138/187)	66.9 (257/384)	53.8 (213/396)
Any combination of oral LLM (≥2 drugs)	12.7 (499/3,928)	0.0 (0/124)	1.3 (3/223)	0.0 (0/169)	10.7 (20/187)	10.5 (41/392)	18.8 (75/398)
PCSK9 inhibitor combined with any oral agent	0.4 (17/3,928)	0.0 (0/124)	0.0 (0/223)	0.0 (0/169)	0.0 (0/187)	0.5 (2/392)	1.8 (7/398)

Values are % (n/N), mean ± SD, or median (Q1-Q3). To convert cholesterol values from mmol/L to mg/dL multiply by 38.67. ^aMedian lipoprotein (a) [Lp(a)] levels across World Health Organization regions were the following: Africa (AFR): 62 nmol/L (Q1-Q3: 30-115 nmol/L), Americas (AM): 40 nmol/L (Q1-Q3: 14-181 nmol/L), Eastern Mediterranean (EM): 27 nmol/L (Q1-Q3: 13-62 nmol/L), Europe (EUR): 35 nmol/L (Q1-Q3: 11-132 nmol/L), Southeast Asia (SEA): 26 nmol/L (Q1-Q3: 8-83 nmol/L) and Western Pacific (WP): 22 nmol/L (Q1-Q3: 9-65 nmol/L).

ApoB = apolipoprotein B; CKD = chronic kidney disease; FH = familial hypercholesterolemia; HDL-C = high-density lipoprotein cholesterol; LDL-C = Low-density lipoprotein cholesterol; LLM = Lipid-lowering medication; Lp(a) = lipoprotein (a); PCSK9 = proprotein convertase subtilisin/kexin type 9; UAE = United Arab Emirates.

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participants had Lp(a) measurements obtained for this analysis. Blood draws were successfully conducted in 97.2% of cases, and 52.4% of participants were classified as having an MI (ST-segment elevation MI/non-ST-segment elevation MI) as their recruiting event. Participants with available Lp(a) measurements had similar characteristics to those of the overall INTERASPIRE cohort ([Supplemental Table 1](#)).

Subjects' characteristics overall and by country are presented in [Table 1](#). The majority were men (78.9%) and mean age was 60.2 ± 10.2 years. Almost one-half of the participants reported having been diagnosed with diabetes (44.0%), 9.7% with familial hypercholesterolemia, and 9.5% with CKD. At study interview, the majority were on statin monotherapy (23.1% low/moderate and 50.7% high intensity), 12.7% received any combination of oral lipid-lowering medication (LLM), and 0.4% were treated with PCSK9-targeted therapy; 468 (11.9%) participants did not receive any LLM. Participants' characteristics differed across countries. African countries had the highest proportions of enrolled women (ranging from 30.6%-49.7%), CKD was more prevalent in the Philippines (14.7%) and Poland (13.7%), whereas the prevalence of diabetes was the highest in the UAE (69.3%) and Kenya (67.7%). LLM use also varied significantly by country.

The median Lp(a) concentration in the overall study population was 32 nmol/L (Q1-Q3: 11-89 nmol/L). Marked variations in Lp(a) levels were observed between regions and between countries within the same region ([Table 1](#)). By region, the highest

Lp(a) levels were observed in Africa (62 nmol/L [Q1-Q3: 30-115 nmol/L]) and the lowest in Western Pacific (22 nmol/L [Q1-Q3: 9-65 nmol/L]; $P < 0.0001$). Within Europe, Lp(a) levels were higher in Portugal (59 nmol/L [Q1-Q3: 21-150 nmol/L]) compared with Poland (19.5 nmol/L [Q1-Q3: 8-89.5 nmol/L]; $P < 0.0001$). In South America, levels were higher in Colombia (46 nmol/L [Q1-Q3: 15-122 nmol/L]) compared with Argentina (32 nmol/L [Q1-Q3: 11-99 nmol/L]; $P = 0.025$). In the Western Pacific, levels in Malaysia (39.5 nmol/L [Q1-Q3: 14.5-95 nmol/L]) exceeded those observed in the other countries from that region (eg, Philippines 14 nmol/L [Q1-Q3: 7-40 nmol/L]; $P < 0.0001$). Violin plots graphically illustrate the skew in the distribution of Lp(a) between countries grouped by WHO region with evidence of significant variability between and within regions ([Figure 1](#)). Notably, the distributions of Lp(a) in Colombia and Portugal resembled that of the 3 countries from Africa.

Lp(a) concentration by demographics, comorbidities and lipid-lowering therapy are presented in [Supplemental Figure 1](#). Women had higher crude Lp(a) levels than men (44 nmol/L [Q1-Q3: 16-111 nmol/L] vs 29 nmol/L [Q1-Q3: 10-83 nmol/L]; $P < 0.0001$), but no significant differences were found between age groups ($P = 0.11$). The crude odds of having Lp(a) ≥125 nmol/L were higher in women than in men (OR: 1.36, 95% CI: 1.12-1.64). This association remained unchanged after adjusting for age, smoking, systolic BP, diabetes, and eGFR, yielding an

TABLE 1 Continued						
EUR		SEA	WP			
Poland (n = 288)	Portugal (n = 251)	Indonesia (n = 398)	China (n = 305)	Malaysia (n = 400)	Philippines (n = 397)	Singapore (n = 396)
27.8 (80/288)	23.5 (59/251)	14.1 (56/398)	19.0 (58/305)	17.0 (68/400)	22.9 (91/397)	7.1 (28/396)
63.9 ± 9.6	63.7 ± 9.3	58.0 ± 8.7	60.1 ± 9.9	58.5 ± 10.8	58.2 ± 9.9	60.6 ± 9.0
5.3 (15/283)	4.5 (11/244)	3.6 (12/335)	8.7 (25/288)	6.6 (24/361)	19.0 (71/374)	19.3 (63/327)
36.1 (104/288)	32.7 (82/251)	32.4 (129/398)	43.6 (133/305)	48.8 (195/400)	46.6 (185/397)	46.7 (185/396)
13.7 (39/285)	11.3 (28/247)	6.1 (24/391)	6.3 (19/301)	12.4 (48/387)	14.7 (57/389)	7.4 (29/391)
19.5 (8-89.5)	59 (21-150)	26 (8-83)	22 (7-70)	39.5 (14.5-95)	14 (7-40)	21 (9-60)
2.04 ± 0.90	1.95 ± 0.87	2.69 ± 0.95	1.65 ± 0.76	2.64 ± 1.09	2.22 ± 1.08	2.05 ± 0.82
2.65 ± 0.99	2.53 ± 1.06	3.46 ± 1.17	2.28 ± 0.92	3.32 ± 1.22	2.91 ± 1.34	2.65 ± 0.96
0.77 ± 0.24	0.76 ± 0.28	—	—	0.92 ± 0.31	0.79 ± 0.31	0.76 ± 0.24
14.2 (41/288)	10.8 (27/251)	70.0 (278/397)	69.2 (211/305)	8.0 (32/400)	7.1 (28/392)	5.8 (23/396)
59.0 (170/288)	35.5 (89/251)	17.6 (70/397)	2.6 (8/305)	77.3 (309/400)	73.0 (286/392)	62.4 (247/396)
16.3 (47/288)	41.8 (105/251)	1.8 (7/398)	17.4 (53/305)	5.5 (22/400)	3.8 (15/397)	28.0 (111/396)
0.0 (0/288)	0.0 (0/251)	0.0 (0/398)	1.6 (5/305)	0.0 (0/400)	0.0 (0/397)	0.8 (3/396)

adjusted OR of 1.31 (95% CI: 1.07-1.59). Lp(a) levels were similar irrespective of the type of CHD presentation (eg, MI vs PCI vs CABG) and the time from the recent CHD event (Supplemental Figure 1). Lp(a) levels were higher in participants who reported concomitant PAD (82 nmol/L [Q1-Q3: 17-119 nmol/L] vs 31 nmol/L [Q1-Q3: 11-88 nmol/L]; $P = 0.034$). Lp(a) levels also increased progressively with declining eGFR (eGFR ≥ 90 mL/min/1.73 m²: 28 nmol/L [Q1-Q3: 10-83 nmol/L], eGFR 60-90 mL/min/1.73 m²: 33 nmol/L [Q1-Q3: 11-88 nmol/L], eGFR 30-60 mL/min/1.73 m²: 37 nmol/L [Q1-Q3: 13-110 nmol/L], and eGFR < 30 mL/min/1.73 m²: 48 nmol/L [Q1-Q3: 16-113 nmol/L]; $P = 0.0002$).

Overall, the proportions of patients with Lp(a) levels ≥ 125 , 150, 175, and 200 nmol/L were 17.6%, 13.0%, 9.3%, and 6.2%, respectively (Figure 2A). The proportions of patients above the Lp(a) inclusion thresholds for ongoing cardiovascular outcome trials showed substantial variation across countries (Figure 2B). The highest proportions of patients with Lp(a) ≥ 150 , 175, and 200 nmol/L were observed in Portugal (25.5%, 18.3%, and 11.6%) and Kenya (21.0%, 15.3%, and 8.9%), whereas the lowest rates were found in the UAE (6.0%, 3.3%, and 1.0%) and Philippines (4.3%, 2.5%, and 1.3%).

The associations between increasing Lp(a) and other lipids were markedly nonlinear, including a consistently flat slope for Lp(a) concentrations between 50 and 200 nmol/L. However, as expected, the proportion of Lp(a) particles relative to total apoB-containing lipoproteins increased linearly throughout the entire concentration range (locally estimated scatterplot smoothing plots) (Figure 3). Table 2 shows the distribution of lipids across deciles of Lp(a). Across Lp(a) deciles, Lp(a) increased from 3.0

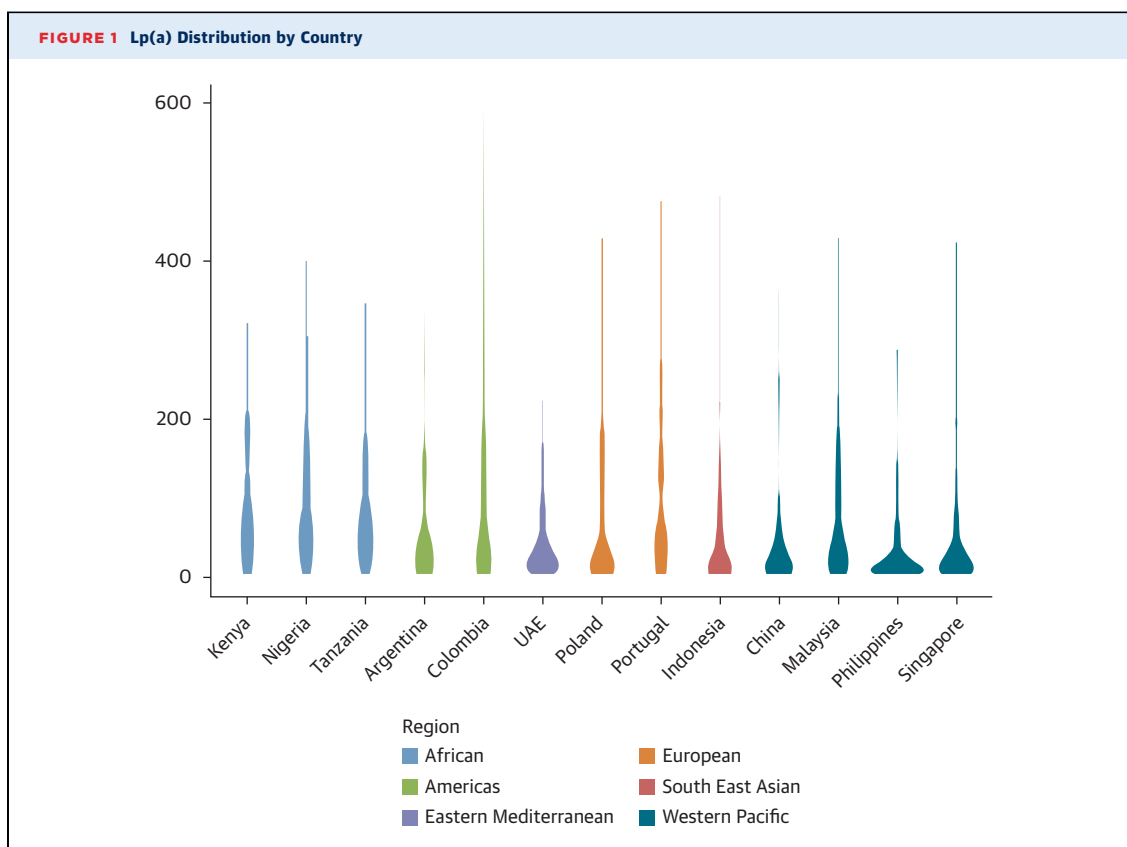
to 211.0 nmol/L, LDL-C from 1.84 to 2.22 mmol/L, non-HDL-C from 2.51 to 2.91 mmol/L, and total ApoB from 0.71 to 0.88 g/L. VLDL-C concentrations remained constant and the proportion of Lp(a) particles to total ApoB particles rose from 0.2% to 12.6%. No major differences were observed in the use of lipid-lowering therapies across Lp(a) deciles (Supplemental Table 2).

Comparing deciles 1 to 3, Lp(a) levels increased modestly, from 3 to 11 nmol/L, with corresponding absolute increases in LDL-C by 0.13 mmol/L, non-HDL-C by 0.09 mmol/L, ApoB by 0.02 g/L, and Lp(a)/ApoB ratio by 0.6% (from 0.2% to 0.8%). By contrast, deciles 8 to 10 showed a marked increase in Lp(a) levels, rising by 122 nmol/L (from 89 to 211 nmol/L). LDL-C and non-HDL-C concentrations remained relatively stable, while ApoB concentrations increased by 0.04 g/L. The Lp(a)/ApoB ratio demonstrated an absolute increase of 7.0% (from 5.6% to 12.6%).

Among participants with Lp(a) ≥ 125 nmol/L at the time of the interview, 14.0% were smokers, and 60.6% did not report regular physical activity. The mean BMI of study participants was 27.3 ± 4.9 kg/m², with 67.9% having BMI ≥ 25.0 kg/m². Mean systolic and diastolic BP levels were 129.6 ± 20.6 mm Hg and 78.6 ± 11.5 mm Hg, respectively, with 57.5% having BP $\geq 130/80$ mm Hg. The mean LDL-C was 2.51 ± 1.11 mmol/L, with 89.4% having LDL-C ≥ 1.4 mmol/L. Among patients with diabetes, the mean HbA1c was $7.44\% \pm 2.0\%$ and 44.8% had HbA1c $\geq 7\%$.

DISCUSSION

Leveraging data from the INTERASPIRE study, the present analyses provide additional information



about the distribution of Lp(a) levels in patients with CHD across geographical regions, with immediate implications for novel Lp(a)-lowering therapies currently under development (**Central Illustration**). First, median Lp(a) levels (32 nmol/L) with 82.4% of patients below the threshold typically associated with increased ASCVD risk, suggest that the overall contribution of Lp(a) to CHD at the population level may be no more than most other cardiovascular risk factors. Second, depending upon the Lp(a) thresholds used in ongoing cardiovascular outcome trials, the proportion of patients eligible for different emerging Lp(a)-lowering therapies varies and may range from 6.2% (≥ 200 nmol/L) to 13.0% (≥ 150 nmol/L) globally, with considerable variation in the proportion of eligible patients both between WHO regions, but also between countries within the same region. Third, LDL-C, non-HDL-C, and total ApoB were not meaningfully predictive of Lp(a) concentration across the majority of the Lp(a) range, meaning that Lp(a) itself must be measured if we are to identify those with elevated levels. Finally, most patients with elevated Lp(a) had poorly controlled levels of traditional risk factors, which means if emerging Lp(a)-lowering therapies are unavailable in some countries, traditional risk

factor management needs to improve several fold to offset the residual risk from high Lp(a).

The vast majority of epidemiological and genetic studies have been conducted in populations of European ancestry without prior ASCVD.²²⁻²⁶ These have shown that elevated Lp(a) concentrations exceeding ~ 125 nmol/L are associated with a 2- to 3-fold increase in risk of ASCVD across a range of phenotypes including MI, ischemic stroke, and PAD, as well as being associated with aortic valve stenosis and cardiovascular and all-cause mortality.¹⁻³ These elevated levels are not uncommon, affecting approximately 1 in 5 individuals in the general population,¹⁻³ and coupled with emerging evidence, suggesting that Lp(a) carries a 6.6-fold higher ASCVD risk than LDL when analyzed per ApoB particle.¹¹ Collectively, these observations have led to speculation about whether Lp(a) contributes to a higher population-attributable risk globally than other traditional risk factors,¹¹ and have hastened the search for novel therapies that meaningfully reduce Lp(a).^{27,28}

Previous studies have shown that Lp(a) levels vary by ancestry with the lowest concentrations typically observed in individuals from East Asian, European, and Southeast Asian regions; intermediate levels in Latin America, the Middle East, and South Asia; and

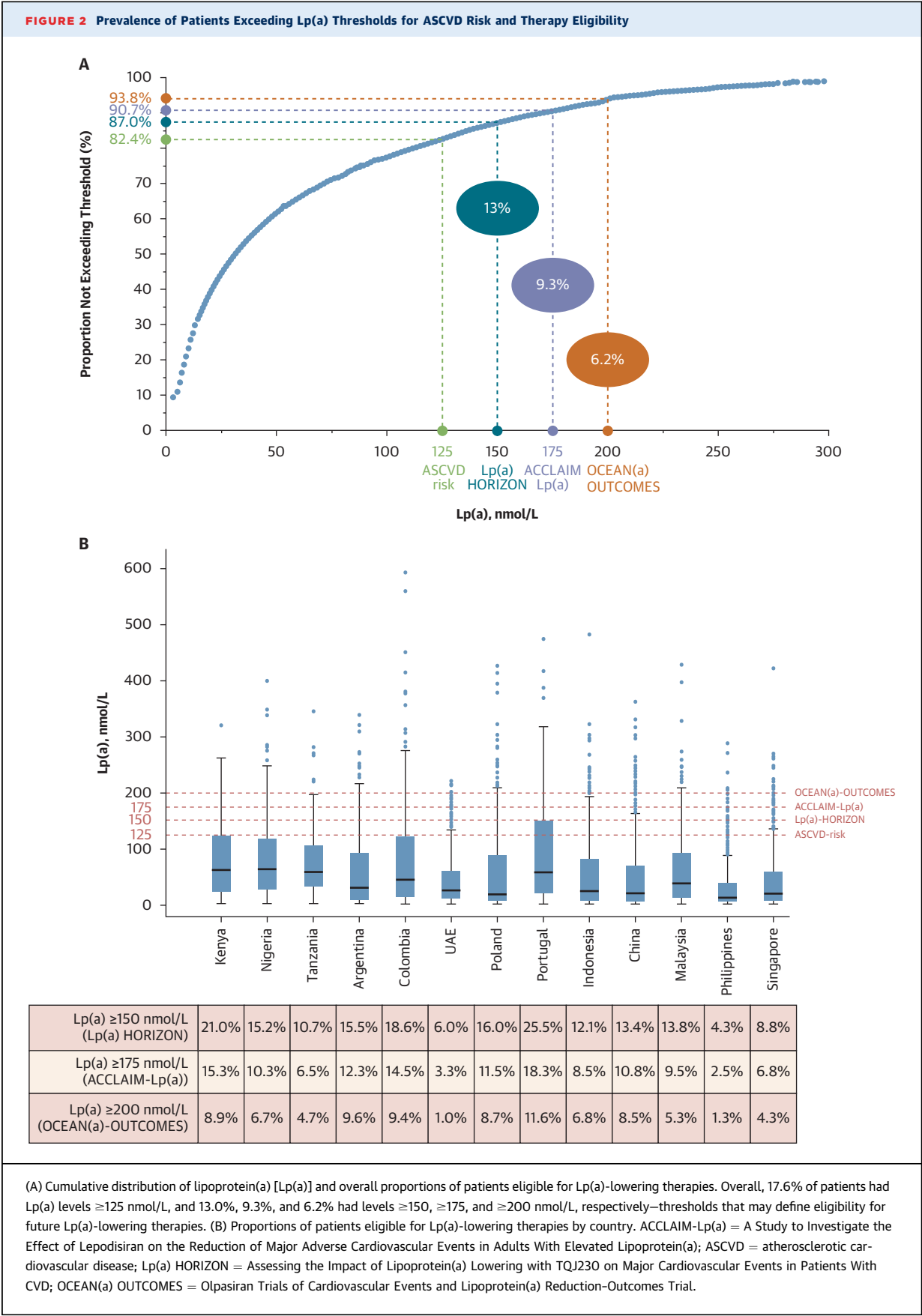
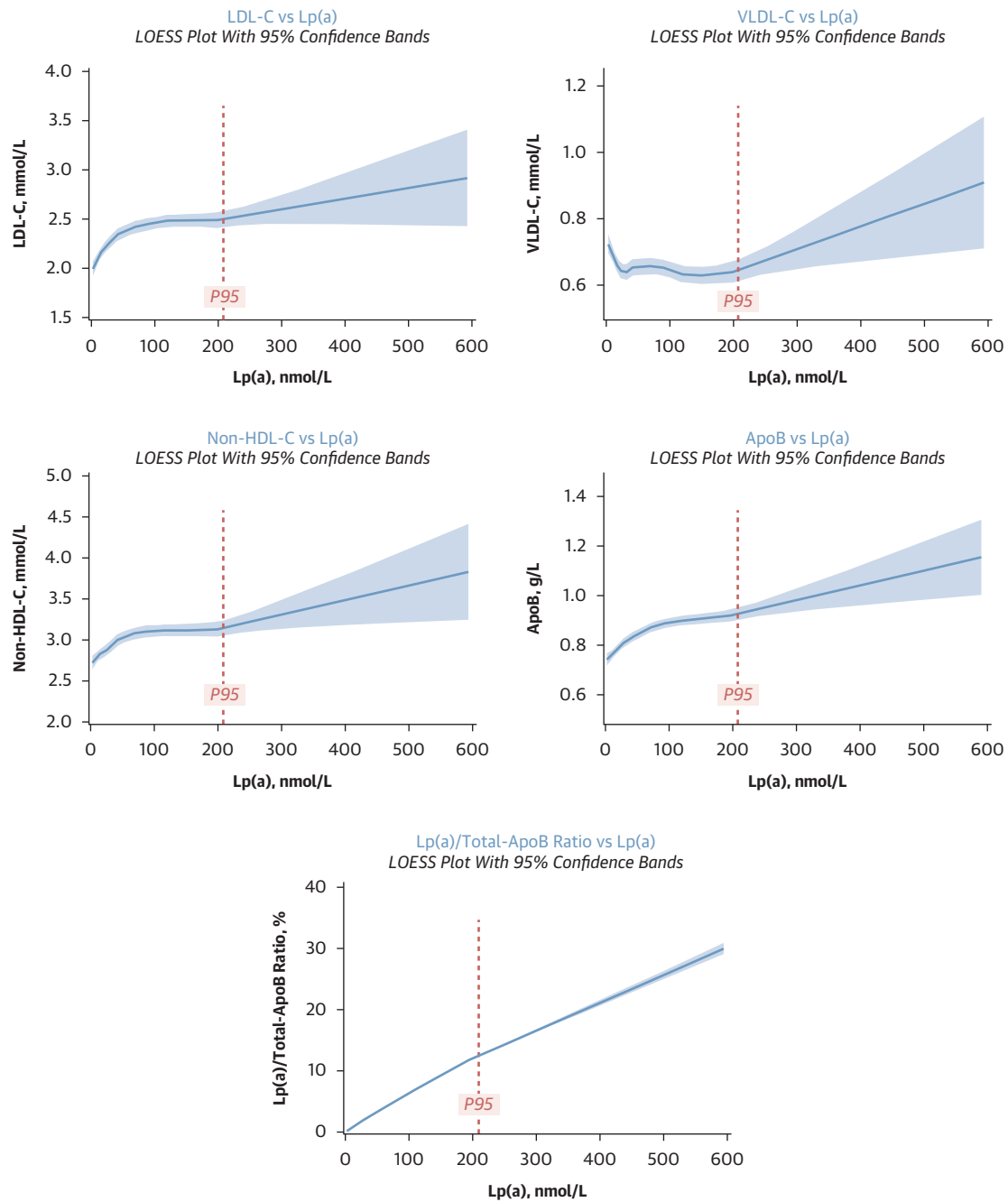


FIGURE 3 LOESS Plots for the Associations Between Various Lipid Measures and Lipoprotein(a)



Vertical line represents the 95th percentile. To convert cholesterol values from mmol/L to mg/dL, multiply by 38.67. ApoB = apolipoprotein B; LDL-C = low-density lipoprotein cholesterol; LOESS = locally estimated scatterplot smoothing; Lp(a) = lipoprotein(a); non-HDL-C = non-high-density lipoprotein cholesterol; VLDL-C = very-low-density lipoprotein cholesterol.

TABLE 2 Lipid Concentrations Across Lp(a) Deciles

	Lp(a) Deciles										
	Total	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Lp(a), nmol/L	32 (11-89)	3 (3-3)	7 (6-8)	11 (10-13)	17 (15-19)	26 (23-29)	39 (35-42)	58 (52-65)	89 (81-100)	137 (125-151)	211 (189-253)
LDL-C, mmol/L	2.06 (1.56-2.78)	1.84 (1.42-2.47)	1.88 (1.44-2.41)	1.97 (1.53-2.59)	1.97 (1.54-2.78)	1.95 (1.50-2.70)	2.16 (1.59-2.82)	2.21 (1.60-2.94)	2.27 (1.70-3.00)	2.22 (1.70-3.10)	2.22 (1.78-2.95)
VLDL-C, mmol/L	0.56 (0.44-0.75)	0.59 (0.46-0.81)	0.56 (0.45-0.75)	0.58 (0.45-0.78)	0.57 (0.45-0.77)	0.54 (0.43-0.71)	0.56 (0.42-0.72)	0.56 (0.44-0.76)	0.55 (0.44-0.77)	0.54 (0.42-0.71)	0.56 (0.44-0.75)
Non-HDL-C, mmol/L	2.69 (2.11-3.51)	2.51 (1.98-3.29)	2.46 (2.01-3.04)	2.60 (2.06-3.47)	2.63 (2.09-3.56)	2.53 (2.07-3.40)	2.77 (2.10-3.62)	2.85 (2.14-3.65)	2.89 (2.25-3.70)	2.83 (2.21-3.74)	2.91 (2.29-3.67)
Total ApoB, g/L	0.77 (0.62-0.97)	0.71 (0.57-0.90)	0.70 (0.57-0.83)	0.73 (0.59-0.92)	0.73 (0.58-0.96)	0.72 (0.61-0.94)	0.76 (0.62-0.96)	0.80 (0.64-0.98)	0.84 (0.68-1.04)	0.85 (0.68-1.06)	0.88 (0.74-1.08)
Lp(a)/total ApoB (nmol/L) ratio, %	2.2 (0.8-5.6)	0.2 (0.2-0.3)	0.5 (0.4-0.6)	0.8 (0.6-0.1)	1.2 (0.9-1.5)	1.8 (1.4-2.3)	2.6 (2.1-3.2)	3.9 (3.0-4.7)	5.6 (4.4-6.9)	8.4 (6.4-10.4)	12.6 (10.2-16.5)
Values are medians (Q1-Q3). To convert cholesterol values from mmol/L to mg/dL, multiply by 38.67. Total ApoB (nmol/L) = (Total ApoB [g/L] × 100) × 19.49. VLDL-C = very-low-density lipoprotein cholesterol; other abbreviations as in Table 1 .											

the highest levels in sub-Saharan Africa.^{5,28,29} Similarly, in our analysis, the highest Lp(a) levels were noted in African countries, and the lowest most skewed distributions in Western Pacific countries. Beyond regional variability, our study highlights marked differences between countries within the same region likely reflecting migration patterns and genetic admixture, as seen in Colombia and Portugal, where Lp(a) distributions resembled African countries. These countries have a significant population originating from the Caribbean and Angola, respectively, which potentially explains our findings. Likewise, participants from Malaysia with a significant population from South Asia exhibited higher Lp(a) levels compared with neighboring countries like the Philippines and Singapore, which had the lowest Lp(a) levels in our study.

The advent of therapies targeting apo(a) production, such as antisense oligonucleotides and small interfering RNA therapies, offers the potential to lower Lp(a) levels by over 80%^{27,28} and are currently being investigated in cardiovascular outcome trials using single Lp(a) thresholds irrespective of ancestry, ranging from ≥150, 175, to 200 nmol/L.^{12,13,18} Our study suggests that a patient with CHD and a Lp(a) ≥150 nmol/L from Nigeria or Colombia, for instance, represents an entirely different percentile of Lp(a) distribution within their country than someone in the Philippines or Singapore with CHD and the same Lp(a) level. In the INTERHEART study, although the same Lp(a) exposure (>50 mg/dL or ~125 nmol/L) was not associated with increased odds of MI in individuals of African ancestry, it was strongly associated with MI risk in South Asians.⁵ However, a greater proportion of individuals of African ancestry had Lp(a) levels above this threshold

compared with South Asians, meaning that the population-attributable risk varied widely—from 0% in individuals of African ancestry to 9.5% in South Asians.⁵ This underscores the need to better understand how these geographic variations in Lp(a) distribution impact risk prediction if Lp(a) is integrated into risk calculators across different geographical regions as well as the design of future trials of Lp(a)-lowering therapies. Although these epidemiological studies are intriguing, ultimately it remains to be seen whether the ongoing Lp(a) cardiovascular outcome trials will demonstrate similar cardiovascular benefits in different ethnic groups and across different countries.

If ongoing Lp(a)-lowering trials are positive, then our observations raise concerns about equitable access to these treatments, particularly in regions with a high burden of elevated Lp(a), such as Africa and Colombia, where access may be limited despite potential significant need. Disparities in insurance coverage within countries like the United States could disproportionately affect Black and South Asian populations having higher Lp(a) levels, exacerbating health inequities, as already highlighted in previous multiethnic cohort studies.³⁰⁻³² Another example is UAE, where differences in eligibility criteria based on proportions of patients above treatment thresholds will mean that 6.0% of patients might be eligible for pelacarsen (6.0%) vs 1.0% for olpasiran, limiting physician and patient choice. Additionally, the lower Lp(a) levels observed in the Western Pacific raises a dilemma about whether Lp(a) thresholds for access to therapy should be adjusted for these countries, or new trials with lower Lp(a) entry criteria will be needed to provide evidence of efficacy across a broader Lp(a) range in different countries.

CENTRAL ILLUSTRATION Global Variation in Lipoprotein(a) Levels Among Patients With Coronary Heart Disease From the INTERASPIRE Study

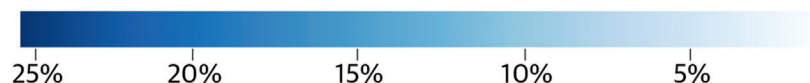
Purpose: To assess variation in Lp(a) levels among coronary heart disease (CHD) patients across the globe and estimate eligibility for future Lp(a)-lowering therapies



Eligibility rates for Lp(a)-lowering therapies across regions/countries

Africa-Kenya	21.0%	15.3%	8.9%
Africa-Nigeria	15.2%	10.3%	6.7%
Africa-Tanzania	10.7%	6.5%	4.7%
Americas-Argentina	15.5%	12.3%	9.6%
Americas-Colombia	18.6%	14.5%	9.4%
Europe-Poland	16.0%	11.5%	8.7%
Europe-Portugal	25.5%	18.3%	11.6%
Southeast Asia-Indonesia	12.1%	8.5%	6.8%
Eastern Mediterranean-UAE	6.0%	3.3%	1.0%
Western Pacific-China	13.4%	10.8%	8.5%
Western Pacific-Malaysia	13.8%	9.5%	5.3%
Western Pacific-Philippines	4.3%	2.5%	1.3%
Western Pacific-Singapore	8.8%	6.8%	4.3%

Lp(a) ≥ 150 nmol/L (Lp(a)HORIZON) Lp(a) ≥ 175 nmol/L (ACCLAIM-Lp(a)) Lp(a) ≥ 200 nmol/L (OCEAN(a)-OUTCOMES)



Median Lp(a) levels

Median Lp(a) levels overall: 32 nmol/L (11-89)

17.6% having levels ≥ 125 nmol/L

Eligibility rates for Lp(a)-lowering therapies

13%

Lp(a) ≥ 150 nmol/L (Lp(a)HORIZON)

9.3%

Lp(a) ≥ 175 nmol/L (ACCLAIM-Lp(a))

6.2%

Lp(a) ≥ 200 nmol/L (OCEAN(a)-OUTCOMES)

Median Lp(a) in CHD patients is below common risk thresholds and varies by region and country which may impact equitable access to Lp(a)-lowering therapies.

Barkas F, et al. JACC. 2025;85(21):2028-2042.

ACCLAIM-Lp(a) = A Study to Investigate the Effect of Lepodisiran on the Reduction of Major Adverse Cardiovascular Events in Adults With Elevated Lipoprotein(a); CHD = coronary heart disease; HORIZON = Assessing the Impact of Lipoprotein(a) Lowering with TQJ230 on Major Cardiovascular Events in Patients With CVD; Lp(a) = lipoprotein (a); OCEAN(a) OUTCOMES = Olpasiran Trials of Cardiovascular Events and Lipoprotein(a) Reduction-Outcomes Trial.

There are similarities and significant differences between the present study and prior relevant studies that merit discussion. The INTERHEART study was a large, standardized case-control study of MI which provided the first estimates of global variation in Lp(a) levels in those with and without MI focusing on ethnicity rather than country, by pooling data from 52 countries.⁵ Using an isoform-insensitive assay, rather than the current gold-standard molar assay, median Lp(a) levels among the cases of first MI varied substantially across ethnic groups ranging from 7.8 mg/dL in Chinese to 27.2 mg/dL in Africans and were only modestly higher than control subjects.⁵ By contrast, Lp(a)HERITAGE was a pragmatic medical record-based registry across 48 countries which did not involve a standardized examination of all participants using a consistent operating procedure and equipment.²⁹ Lp(a) measurements were retrieved from medical records from those tested in the previous 5 years, with missing measurements supplemented by additional testing at local laboratories (thereby introducing potential ascertainment bias).²⁹ Although this study did demonstrate geographical variations in Lp(a), the use of local laboratory measurements in mg/dL, which are not apo(a) isoform independent, coupled with laboratory variability between assays used, results in uncertainty about the overall aggregated Lp(a) values reported for different countries.²⁹ That said, these previous studies together with the present study all show that in patients with CHD or ASCVD, Lp(a) levels are well below the 125 nmol/L (~50 mg/dL) thresholds typically considered to be associated with elevated ASCVD risk in general primary prevention populations. Therefore, collectively they infer that the population-attributable risk of Lp(a) for CHD is likely less than some have suggested. For context and comparison, diabetes, which has a global prevalence of ~10%, doubles the risk of CHD, and is present in approximately 25% or more individuals with CHD.³³

Finally, a large, contemporary U.S. retrospective cohort study conducted at 2 academic centers in Boston, suggests that the same Lp(a) exposure may confer differing levels of risk depending upon the presence or absence of ASCVD.⁶ In individuals with pre-existing ASCVD, the risk of cardiovascular events increased with increasing Lp(a) levels but plateaued once Lp(a) exceeded the 70th percentile (≥ 112 nmol/L).⁶ By contrast, in those without prior ASCVD, increased risk was confined to those whose Lp(a) levels exceeded the >90th percentile (≥ 216 nmol/L).⁶ These findings suggest that Lp(a) attributable risk is not binary, but more nuanced and complex depending

upon several factors, including potential differential vulnerability, whereby some individuals (eg, those with established ASCVD) may be more susceptible to the adverse effects of Lp(a) even at lower concentrations.

Our observations of differences in Lp(a) levels in several subgroups with CHD provide further validation of results in other studies. First, we observed that the odds of having Lp(a) ≥ 125 nmol/L was 31% higher in women than in men after adjusting for age and other factors. However, this does not imply a greater causal role of Lp(a) in women, because prior evidence suggests that the same unit difference in Lp(a) is associated with a similar difference in risk of cardiovascular events in both sexes.²³ Previous primary prevention data from the UK Biobank showed that Lp(a) levels were higher in middle-aged women compared with men (22 nmol/L vs 17 nmol/L).²³ Similarly, in the ODYSSEY OUTCOMES (Evaluation of Cardiovascular Outcomes After an Acute Coronary Syndrome During Treatment With Alirocumab) trial, women with recent acute coronary syndrome were older than men and had higher median Lp(a) levels (28.0 mg/dL vs 19.3 mg/dL).³⁴ Second, our findings reaffirm the association of CKD with higher Lp(a) levels.^{1,35,36} Although it was believed that reduced kidney function might impact clearance and hence result in higher Lp(a) levels, recent observations from Mendelian randomization studies suggest that Lp(a) may be a causal factor for CKD.³⁷ This might warrant future Lp(a)-lowering studies in such patients assessing hard renal outcomes. Finally, our observation that CHD participants reporting concomitant PAD had much higher Lp(a) levels than those without PAD is consistent with observations from the ODYSSEY OUTCOMES trial, where higher Lp(a) levels at baseline were associated with a higher risk of PAD events, which was modifiable by alirocumab treatment.³⁸ As proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors lower both LDL-C and Lp(a), it remains unknown whether Lp(a) lowering per se contributed to those benefits. Currently, ACCLAIM-Lp(a) is evaluating major adverse limb events as a secondary outcome of interest in addition to major adverse cardiovascular events.¹⁸

Because Lp(a) carries cholesterol and is an ApoB-containing lipoprotein, we explored whether other lipids predict elevated Lp(a) levels reliably, by comparing their levels across the range of Lp(a). Across much of the range of the Lp(a) distribution, LDL-C, non-HDL-C, VLDL-C, and ApoB remained fairly constant. This means that standard testing of lipid profiles cannot reliably predict elevated Lp(a), underscoring the importance of increasing routine

Lp(a) testing for accurate ASCVD risk assessment, particularly given that Lp(a) is measured in <1% of the population, even in high-income countries.³⁹⁻⁴¹

Overall, Lp(a) accounted for 2.2% of total ApoB-containing lipoproteins, ranging from 0.2% to 12.6% across deciles and increasing with Lp(a) concentration. The relatively constant concentration of total ApoB across the majority of the Lp(a) range suggests a redistribution of the total ApoB-containing lipoproteins at higher Lp(a) levels, probably driven by an increase in apo(a) binding to a fairly constant pool of ApoB. When comparing decile 8 to 10, the median Lp(a) levels were 89 and 211 nmol/L, respectively, with minor differences in LDL-C, non-HDL, or ApoB levels; yet, the proportion of Lp(a) to total ApoB-containing lipoproteins increased significantly, rising from 5.6% to 12.6%. Our findings suggest that if cardiovascular benefits are demonstrated through current trials evaluating Lp(a)-lowering therapies, they are unlikely to result from reductions in cholesterol content or circulating total ApoB-containing lipoproteins. Instead, these benefits are more likely to reflect improvements in other characteristics of the Lp(a) particles. These include their proinflammatory, proatherogenic, and putative antifibrinolytic properties, primarily driven by the significantly high content of oxidized phospholipids carried by apo(a).¹¹ Our decile-based cross-sectional analysis aligns with the results from pelacarsen trials demonstrating only modest reductions in Lp(a)-cholesterol, LDL-C, and ApoB concentration despite 80% lowering of Lp(a).⁴²

Finally, current guidelines recommend at least 1 Lp(a) measurement in adulthood, interpreted in the context of absolute CVD risk, and if Lp(a) levels are elevated, risk factor management should be intensified through lifestyle changes and medications.¹ The recently published analysis of MESA (Multi-Ethnic Study of Atherosclerosis) suggests that optimal control of traditional risk factors may attenuate some of the excess risk associated with elevated Lp(a) in a primary prevention setting.⁴³ If those observations are correct and conventional risk factors can mitigate some of the excess risk associated with elevated Lp(a), then our study highlights to what extent risk factor management would need to be improved among patients with established CHD and elevated Lp(a) levels to mitigate this risk, if investigational therapies for Lp(a) were unavailable. That said, in the present analysis, neither the physicians managing the patients, nor the patients themselves, were aware of the central core laboratory-measured Lp(a) levels. A separate ongoing Lp(a) substudy of INTERASPIRE will further elucidate whether making physicians and

patients aware of their Lp(a) levels influences CHD management by evaluating patients' perceptions of cardiovascular disease risk, the role of Lp(a), and their willingness to adopt behavioral changes, as well as physicians' knowledge, attitudes toward testing, clinical practices, and interest in emerging therapies.

STUDY LIMITATIONS. This study is limited to its cross-sectional analysis and a single Lp(a) measurement, although levels are believed to remain relatively constant over time. A central laboratory was used for 11 of 13 countries, with the remaining 2 countries providing standardized data calibrated to the central laboratory. Data were derived from specific regions and selected hospital sites, limiting generalizability across other countries. However, only 2.1% of the total variance in Lp(a) levels was attributable to differences between sites within countries, 8.8% to differences between countries, and 89.1% to individual variability in our study. Data on ethnicity was not systematically collected and therefore not included in this analysis. Selection and survival bias may also be a factor, because this study retrospectively included patients with a recent CHD event within 6 months to 2 years (or up to 3 years in Africa) and may not fully represent other ASCVD phenotypes or high-risk primary prevention populations, including individuals with subclinical atherosclerosis, defined as the presence of atherosclerotic plaque or vascular changes detected by imaging in the absence of clinical cardiovascular events.⁴⁴ Finally, the male predominance in the INTERASPIRE cohort is similar to prior studies with this design reflecting real-world CHD patterns,^{5,29} because men are more frequently affected and women tend to develop CHD at older ages. That said, the upper age limit of 80 years potentially affected the representation of women in our study, which may limit the generalizability of our findings to women older than age 80 years with CHD.

CONCLUSIONS

The vast majority of patients with CHD have Lp(a) levels that are much lower than what is considered a typical risk-enhancing threshold for ASCVD in general populations, suggesting that the risk attributable to Lp(a) is more complex than previously perceived. The significant geographical variations in Lp(a) levels in CHD patients will likely impact equitable access to emerging therapies should they demonstrate clinical benefit. Finally, elevated Lp(a) cannot be reliably predicted from other lipid parameters, which highlights the need to measuring Lp(a) itself.

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ADDRESS FOR CORRESPONDENCE: Prof Kausik K. Ray, Imperial Centre for Cardiovascular Disease Prevention, Department of Primary Care and Public Health, Imperial College London, 90 Wood Lane, White City Campus, London W12 0BZ, United Kingdom. E-mail: k.ray@imperial.ac.uk.

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APPENDIX For a supplemental figure and tables, please see the online version of this paper.